

Durak Card Game – User Guide

1. Introduction

This application is a 2-player simulation of the traditional card game Durak. In this game, the player competes against a computer-controlled opponent (AI). The main objective of the game is to get rid of all your cards before the opponent. If a player is left with cards at the end, they are considered the loser (also called the “Durak”).

This project was developed as part of the Object-Oriented Programming course and focuses on implementing game logic, AI behavior, and a graphical user interface using Java.

2. How to Run the Application

To run the program, follow these steps:

- Open the project folder in IntelliJ IDEA (or any Java IDE)
- Make sure all dependencies (JavaFX if required) are properly configured
- Locate the main class (DurakForm)
- Run the application

Once the program starts, the game window will open, and you can begin playing.

3. Game Interface Overview

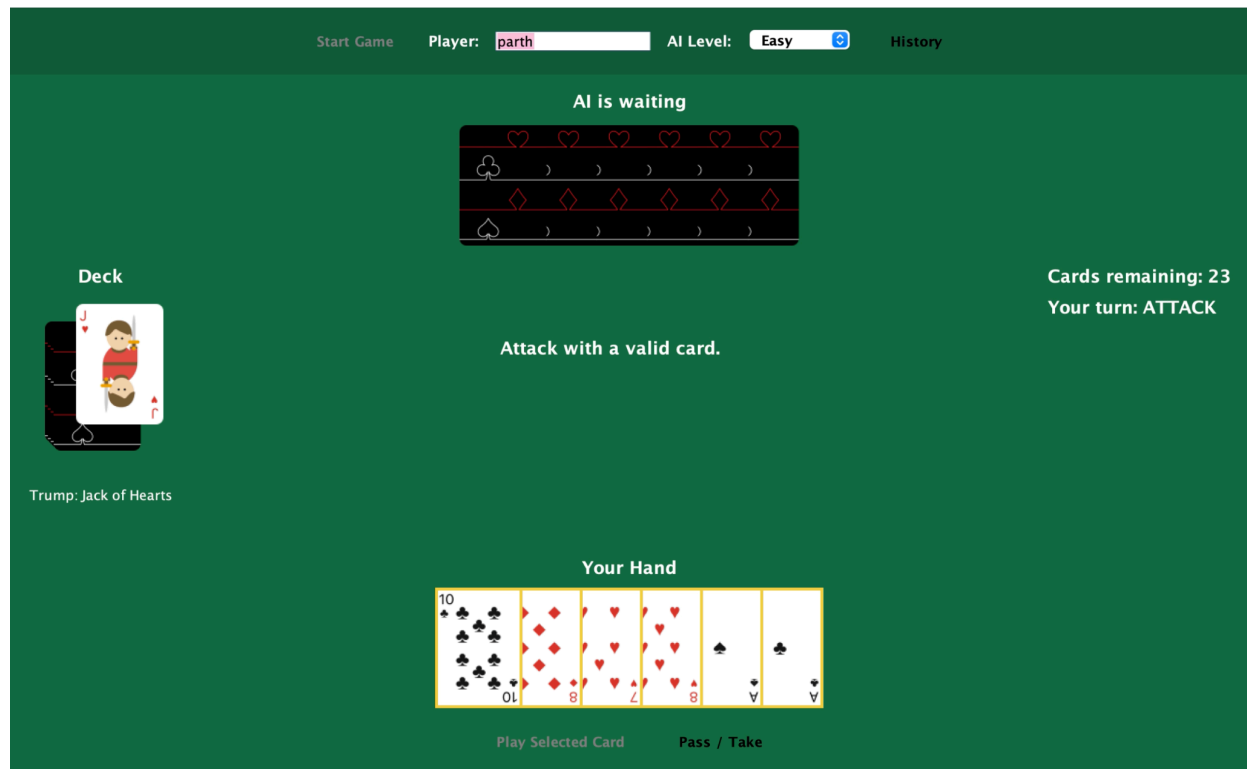
The user interface is designed to be simple and easy to understand:

- At the left of the screen, the trump card is displayed. This determines the strongest suit in the current game.
- The center area shows the table, where attack and defense cards are placed during gameplay.

- At the bottom, the player's hand is displayed as clickable cards.
- Action buttons such as Attack, take and Pass are provided to control gameplay.
- On the right side, the number of cards left and the turn of the player would be displayed.

The interface updates and saves dynamically as the game progresses.

Image from the real game:



4. How to Play the Game

The game is played in turns between the human player and the AI.

- To attack, select a card from your hand and click the "Attack With This Card" button.
- The AI will attempt to defend using a higher card of the same suit or a trump card.
- When the AI attacks, you must defend by selecting a valid card (it will pop up in yellow color) from your hand and clicking "."

- If you cannot defend, you must click “Pass” and take the cards from the table.

The game continues with players alternating turns until one player runs out of cards.

5. Game Rules

- Each player starts with 6 cards.
- A trump suit is selected at the beginning of the game.
- A defending card must be:
 - Higher in rank and same suit, OR
 - Any card from the trump suit (if the attack card is not trump)
- If a player cannot defend, they must pick up all cards on the table.
- The first player to get rid of all cards wins the game.

6. Features Implemented

The application includes several important features:

- A functional 2-player game system (Player vs AI)
- AI opponent with multiple difficulty levels (Easy, Medium, Hard)
- Graphical user interface using JavaFX
- Card images and visual representation of gameplay in the Resources folder.
- Basic game state management (attack, defend, turns)
- Database integration to store user statistics (wins, losses, games played)

7. AI Behavior

The AI in this game is designed with different difficulty levels:

- **Easy Mode:** Plays random valid cards
- **Medium Mode:** Uses simple logic to choose better moves
- **Hard Mode:** Attempts more strategic decisions, such as using trump cards wisely

This allows the player to experience different levels of challenge.

8. Technologies Used

The following technologies were used to develop this project:

- Java (Core programming language)
- JavaFX (Graphical User Interface)
- SQLite (Database for storing user statistics)
- Maven (Build and Dependency Management Tool)
- IntelliJ IDEA (Development environment)

9. Challenges Faced

During the development of this project, several challenges were encountered. These included implementing correct game logic for attack and defense, handling turn-based actions, and ensuring the AI behaves correctly in different situations. Integrating the database and connecting it with the UI was also a challenging part. These issues were resolved through testing, debugging, and improving the code structure.

10. Conclusion

Overall, this project helped in understanding how to apply object-oriented programming concepts such as classes, inheritance, encapsulation, and abstraction in a real-world application like this DURAK Game. It also provided experience in building a complete system that includes logic, UI, and database integration.